

RUNZE SHI

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EDUCATION

University of British Columbia

Vancouver, BC

Master of Applied Science - Mechanical Engineering, GPA 4.10/4.33

Sep. 2024 – Present

University of Wisconsin-Madison

Madison, WI

B.S. Mechanical Engineering, GPA: 3.75/4.0

Sep. 2020 – May. 2024

Core Course: Machine Learning, Design Optimization, Computational Solid Mechanics, Engineering Statistics

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, PyTorch), Git/GitHub, MATLAB, Jupyter, C++ (introductory)

Quantitative Research: Alpha Mining, Cross-Sectional Signal Research, Time-Series Analysis, Hyperparameter Tuning, Out-of-Sample Validation

Statistics & ML: Spearman Correlation, Pearson Correlation, Covariance Matrix, Principal Component Analysis, Random Forest, CNN, LSTM, Reinforcement Learning, Transformers

EXPERIENCE

Celestial Quant Lab | *Quant Researcher-Alpha Mining*

May. 2025 – Present

- Conduct empirical research on equity signals across U.S. and Chinese equity universes, assessing cross-sectional predictive relationships and out-of-sample robustness.
- Designed an iterative Top-K factor-selection procedure over Alpha158-style (Qlib) candidate feature sets, expanding the selected set using RankIC, ICIR, coverage, turnover, and long-short return diagnostics.
- Synthesized signal-evaluation results into research summaries and comparative diagnostics, informing subsequent factor refinement and model-selection decisions.

Laboratory for Structural Efficiency, UBC | *Research Assistant*

Sep. 2024 – Present

- Developed high-fidelity multi-physics simulation pipelines in Abaqus to model complex transient thermal-mechanical systems and generate large-scale structured datasets for downstream machine learning applications.
- Designed machine-learning informed surrogate models to approximate computationally expensive finite-element simulations, enabling parameter exploration and improved scalability for large-scale engineering tasks.
- Spearheaded a 90%+ acceleration in R&D simulation speed by pioneering inherent strain integration and managing multiphysics data streams to link early-stage thermal behavior to end-state welding distortion.

DJI (Da-Jiang Innovations) | *Data & Intelligent Algorithms Intern*

Jun. 2025 – Sep. 2025

- Developed large-scale Python-based data processing pipelines for 18,000+ samples, including data cleaning, anomaly filtering, label validation, and feature consistency checks to improve downstream model reliability.
- Optimized machine learning classification models through iterative hyperparameter tuning and validation workflows, improving operational prediction accuracy from 82% to over 97%.
- Evaluated multi-model inference workflows across LLMs (e.g., Claude, Gemini) using structured benchmarking metrics including inference accuracy, latency, token efficiency, and robustness under boundary conditions.

SELECTED PROJECTS & LEADERSHIP

Hubble: An End-to-end Alpha Mining System | *Technical Lead & Lead Developer* Feb. 2026 - Present

- Led the development of Hubble, an LLM-driven alpha mining system that automated the full research pipeline from data acquisition and factor generation to evaluation, diagnostic, and iterative refinement across U.S. equities.
- Developed a secure AST factor execution sandbox that constrained LLM-generated formulas to 47 predefined market variables and operators. Rejected overly complex expressions through depth and node-count limits.
- Developed a retrieval-augmented generation (RAG) module and evaluation-feedback loop that used RankIC, ICIR, failed-factor diagnostics, and OOS validation results to guide subsequent factor generation and reduce redundancy.
- Evaluated 100+ alpha candidates on an S&P 500 universe, with selected factors demonstrating positive out-of-sample IC and long-short performance; the best OOS factor achieved RankIC = 0.0245, annualized RankICIR = 1.86, and HAC-significant IC performance ($p < 0.01$).
- Organized the Hubble codebase, experimental artifacts, and evaluation reports into a GitHub repository, enabling reproducible research and contributing to an arXiv technical report on the system design and empirical results: <https://doi.org/10.48550/arXiv.2604.09601>

Adaptive Risk Overlay and OOS Strategy Evaluation Framework | *Lead Developer* May. 2026 - Present

- Designed a modular risk-overlay wrapper for reinforcement-learning trading environments, decoupling policy inference from position-sizing and risk-execution logic to enable configurable, no-retraining ablation studies.
- Developed regime-aware exposure controls using robust volatility diagnostics and calibrated rolling windows.
- Diagnosed and corrected a temporal-alignment issue in historical reference data that invalidated dynamic threshold estimation in early OOS backtests; rebuilt reference selection logic.
- Built automated comparative backtesting and reporting workflows; in a 2022 out-of-sample bear-market evaluation, the selected risk configuration reduced maximum drawdown by 52% relative to the baseline strategy.

Automatic Polar Bear Feeder, UW-Madison Capstone Project | *Team Leader* Sep. 2023 – May. 2024

- Led 4 senior-year students in collaboration with the UW-Madison School of Veterinary Medicine to design and deliver an automated feeding system for Henry Vilas (Madison) and Lincoln Park Zoos (Chicago); successfully reduced polar bear anticipatory behavior and met all complex stakeholder requirements.
- Optimized project timelines by managing vendor relationships and adapting engineering designs to resolve supply chain bottlenecks; bridged communication between technical teams and non-technical stakeholders while directing simulations and testing to validate performance targets.

AWARDS

George R. Werth Scholarship (Mechanical Engineering Department) (\$3000/year)	2023, 2024
UBC International Tuition Award (\$3300/year)	2024, 2025, 2026

HOBBIES

Badminton, Fishing, Painting, Gaming